

Where Is It? Shapes on the Toy Shelf

Answer Key

Grade 1

Quick Answers

- | | |
|---|---|
| 1. (a) shapes on the top shelf = 4, — | 5. (a) shapes next to the small circle = 2, — |
| 2. (a) shapes on the bottom shelf = 3, — | 6. (a) bottom-shelf count after the move = 4, (b) |
| 3. (a) shapes between the circle and the rectangle = 2, — | top-shelf count after the move = 3 |
| 4. > | 7. — |
| | 8. (a) shapes in the whole scene = 7, — |
-

What is verified

2 of 8 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problems 1, 2, 3, 5, 7, 8. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 1

K.G – problems 1, 2, 3, 4, 5, 6, 7, 8

Difficulty 1–4 of 5 across 14 tagged checks.

The shelf, for reference — top: circle, triangle, square, rectangle; bottom: hexagon, small circle, tilted square.

- 1.** (a) How many shapes on the TOP shelf? (b) Farthest LEFT on top?

Solution: (a) Count only the top row: circle, triangle, square, rectangle — four shapes.

4

(b) *Instructor judges:* the first shape on the left is the **circle**.

- 2.** (a) How many shapes on the BOTTOM shelf? (b) Below the triangle?

Solution: (a) Count only the bottom row: hexagon, small circle, tilted square — three shapes.

3

(b) *Instructor judges:* straight down from the triangle sits the **small circle**.

- 3.** (a) How many shapes BETWEEN the circle and the rectangle? (b) NEXT TO the rectangle?

Solution: (a) Between the two end shapes sit the triangle and the square — two shapes.

2

(b) *Instructor judges:* the shape touching distance from the rectangle is the **square**.

4. Top shelf 4 shapes, bottom shelf 3. Write $>$, $<$, or $=$: $4 _ 3$.

Solution: Four is more than three, so

4 > 3

5. (a) How many shapes NEXT TO the small circle? (b) To its RIGHT?

Solution: (a) The small circle has a neighbor on each side: the hexagon on its left and the tilted square on its right — two shapes.

2

(b) *Instructor judges:* the shape to the right is the **tilted square** (accept “square”).

6. The square moves from the top shelf to the bottom shelf. (a) Bottom count then? (b) Top count then?

Solution: (a) The bottom shelf had 3 shapes and gains one: $3 + 1 = 4$.

4

(b) The top shelf had 4 shapes and loses one: $4 - 1 = 3$.

3

7. Finish: the hexagon is $_$ the circle.

Solution: *Instructor judges:* the hexagon sits on the bottom shelf, straight down from the circle, so the sentence is completed with **below** (accept “under” or “beneath”).

8. Challenge: (a) how many shapes in all? (b) Tell where the tilted square is, with two position words.

Solution: (a) Add the shelves: $4 + 3 = 7$ shapes.

7

(b) *Instructor judges:* accept any accurate description using two position words, such as “it is on the BOTTOM shelf, to the RIGHT of the small circle” or “it is BELOW the square and NEXT TO the small circle.”